

Who's to Blame for Survey Instability: Respondents with Nonexistent Preferences or Researchers with Flawed Measures?¹

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Society for Political Methodology, 7/18/2026

¹Slides, paper, data: [GaryKing.org/instability](https://garyking.org/instability)

²Joint work with [Libby Jenke](#), University of Houston

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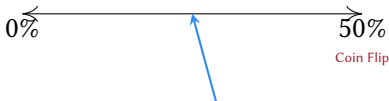
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- **What % answer differently?** 0%  50%
Coin Flip

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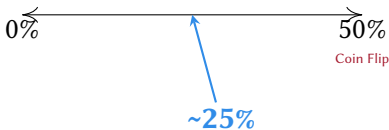
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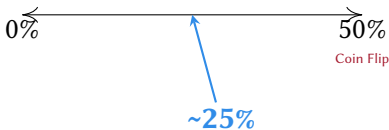
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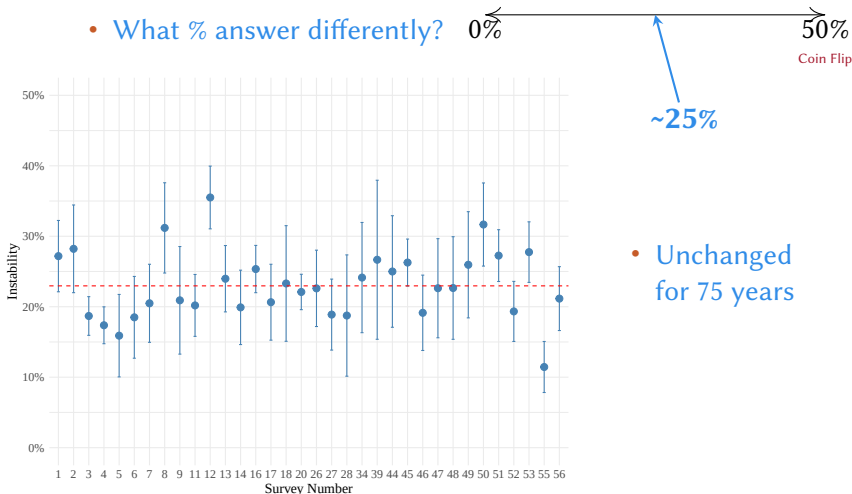
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- **Unchanged**
for 75 years

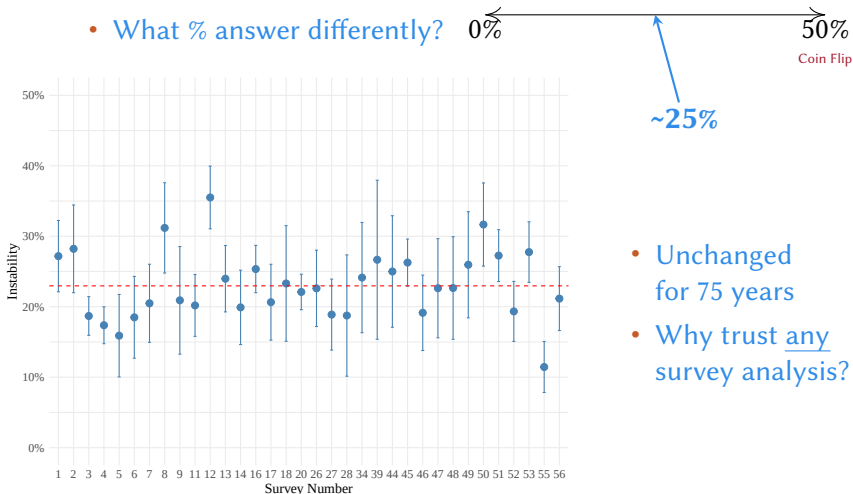
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- Unchanged for 75 years
- Why trust any survey analysis?

Research Strategy

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A Binary Choice Conjoint Question

(Ono and Burden, 2018)

Please carefully review the two potential candidates running for election to the U.S. House of Representatives, detailed below.

	Candidate 0	Candidate 1
Race/Ethnicity	Hispanic	Asian American
Age	52	60
Favorability rating among the public	70%	34%
Position on immigrants	Favors giving citizenship or guest worker status to undocumented immigrants	Opposes giving citizenship or guest worker status to undocumented immigrants
Party affiliation	Republican Party	Democratic Party
Position on abortion	Abortion is not a private matter (pro-life)	Abortion is a private matter (pro-choice)
Position on government deficit	Wants to reduce the deficit through tax increase	Wants to reduce the deficit through tax increase
Salient personal characteristics	Really cares about people like you	Really cares about people like you
Position on national security	Wants to cut military budget and keep the U.S. out of war	Wants to maintain strong defense and increase U.S. influence
Gender	Female	Female
Policy area of expertise	Education	Foreign policy
Family	Single (divorced)	Married (no child)
Experience in public office	12 years	4 years

If you had to choose between them, which of these candidates would you vote to be a member of the U.S. House of Representatives?

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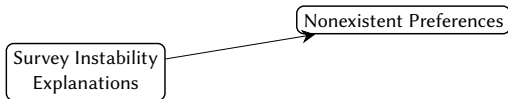
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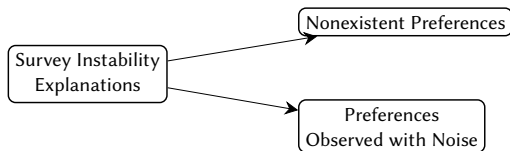
What We Know

Survey Instability
Explanations

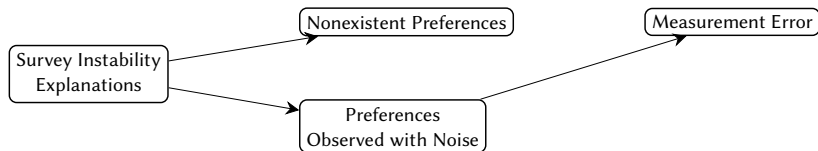
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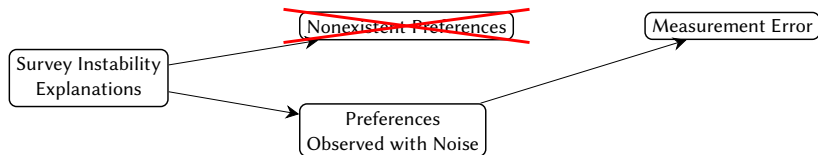
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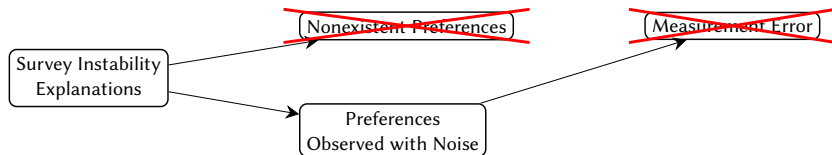
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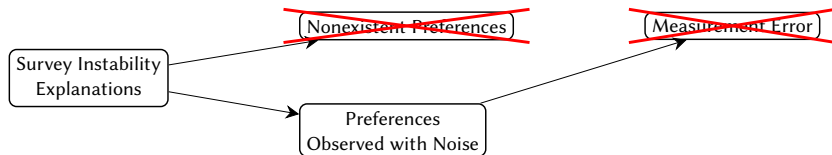
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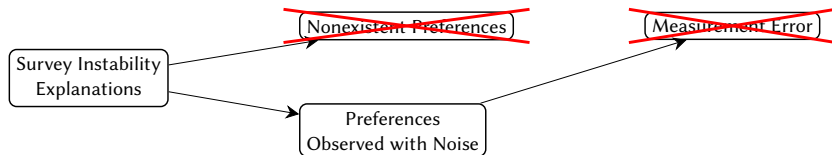


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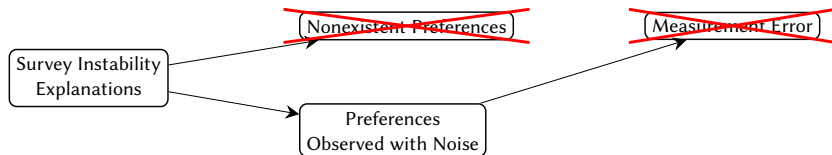
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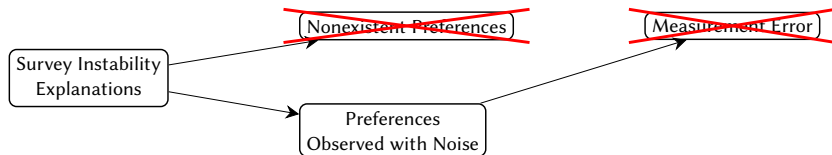
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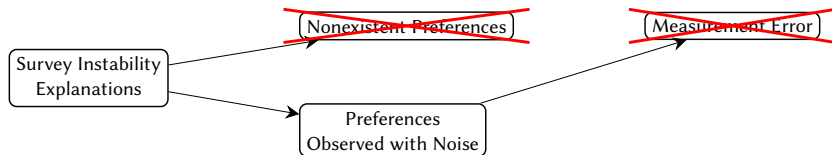
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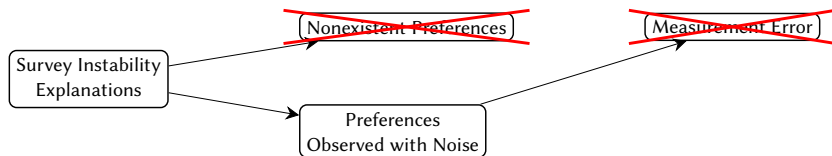
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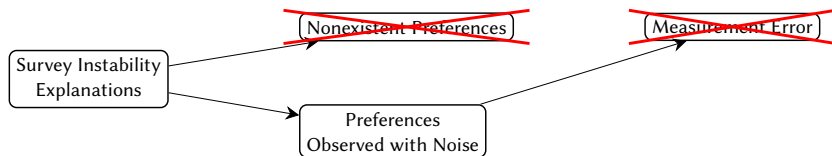
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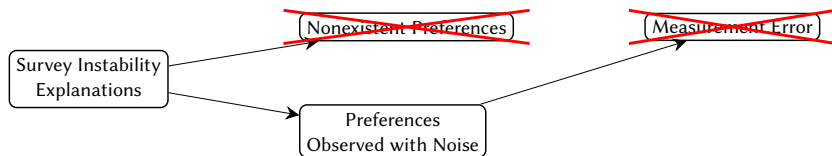
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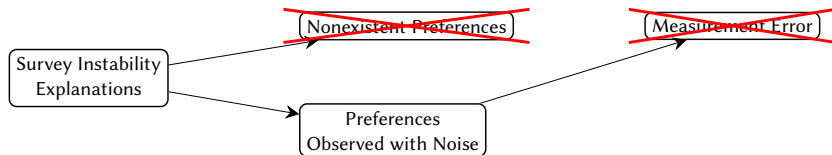


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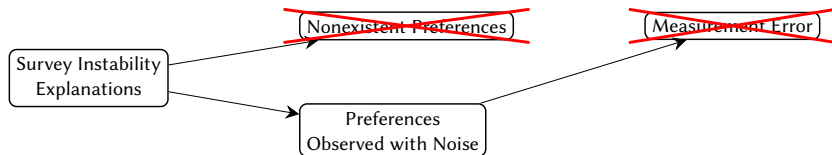
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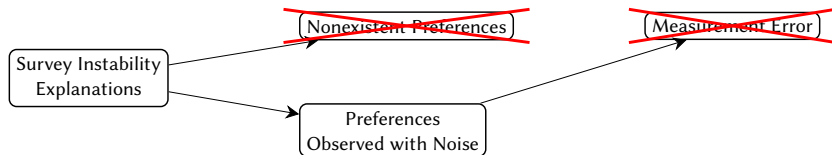
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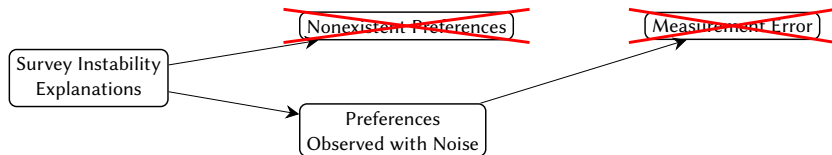
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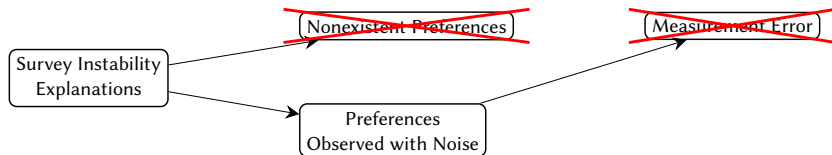
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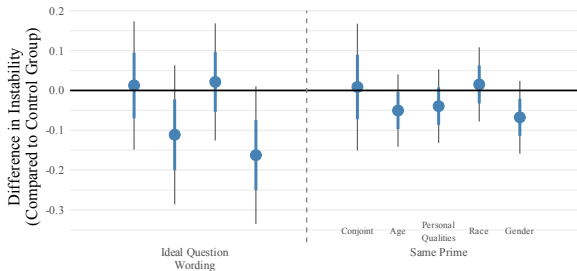
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Reducing Bias Doesn't Reduce Instability



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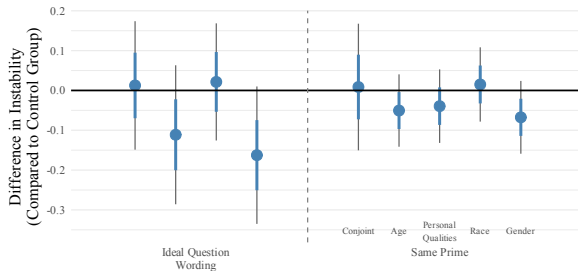
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- Left: 2008 ANES question wording changes

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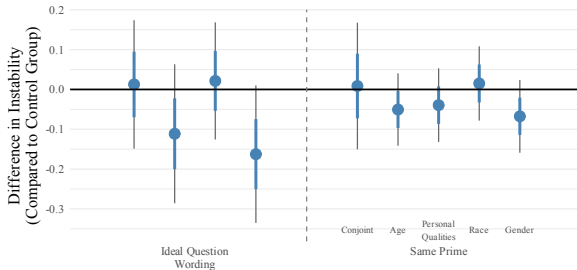
~~Nonexistent Preferences~~

~~Measurement Error~~

Survey Instability

Explains

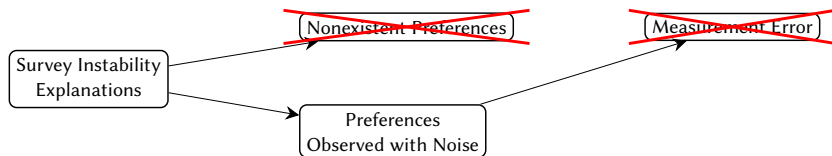
Reducing Bias Doesn't Reduce Instability



- Left: 2008 ANES question wording changes
- Right: different (priming) “considerations”
- reduce bias, not instability



What We Know



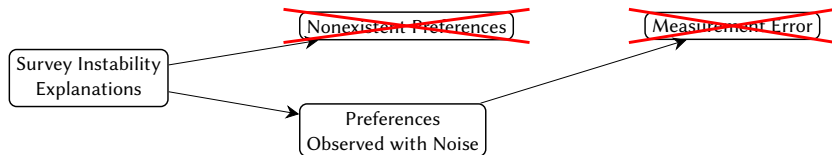
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- ~all evidence contradicts Converse
- Survey findings: numerous, strong, replicable
- **Persuasion attempts:** usually fail
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- $\Rightarrow$ Preferences exist & rarely change (without a reason)

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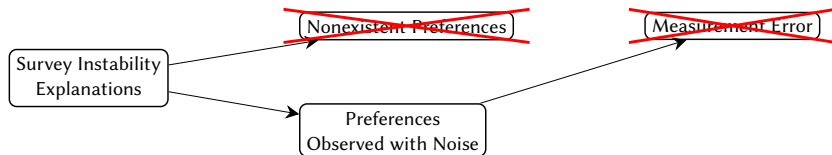
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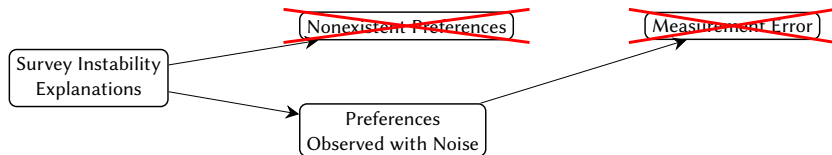
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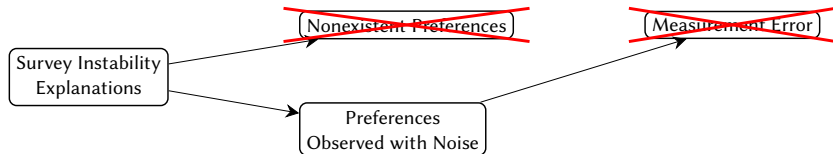
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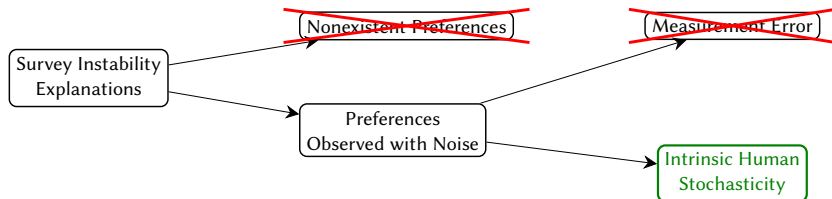
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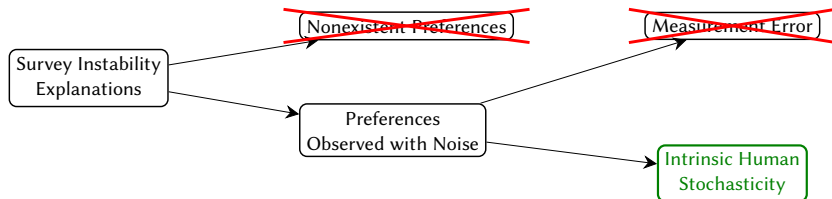
What We Show



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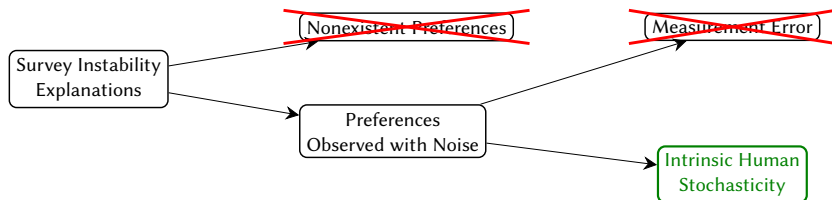


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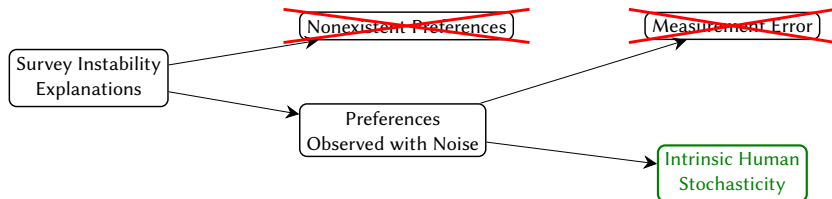
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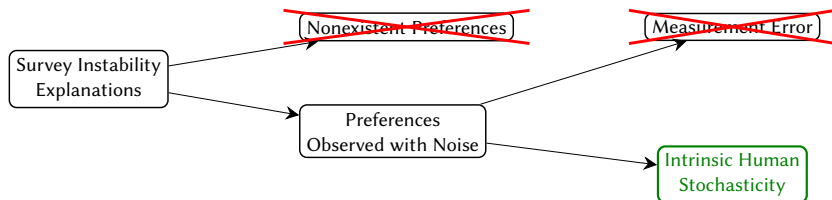
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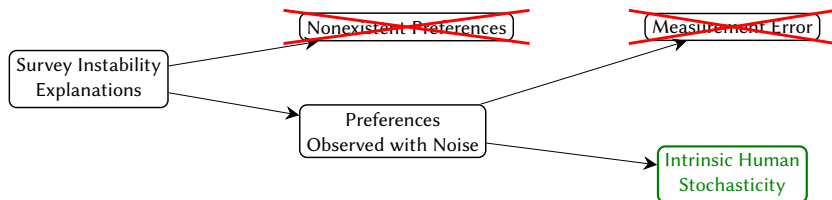
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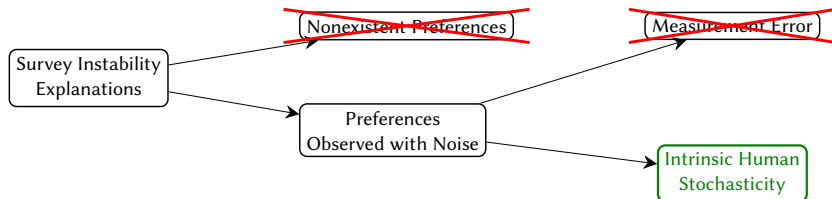
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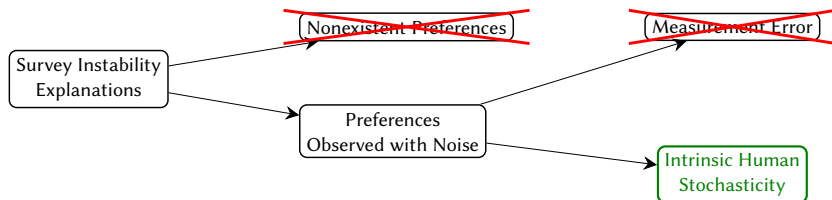
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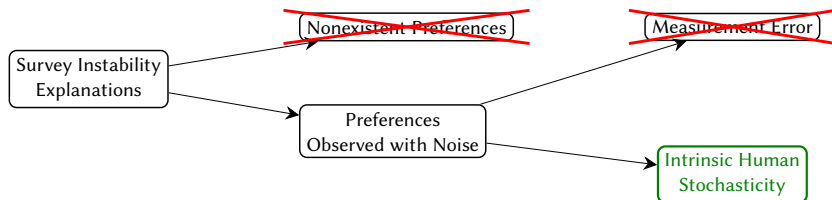
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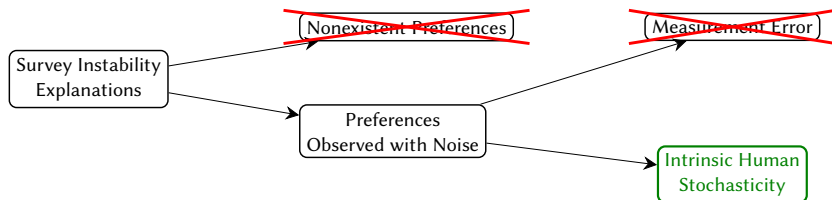
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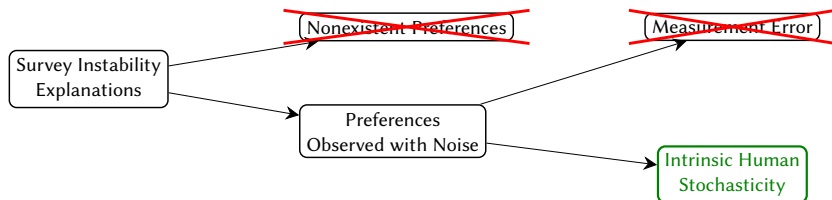
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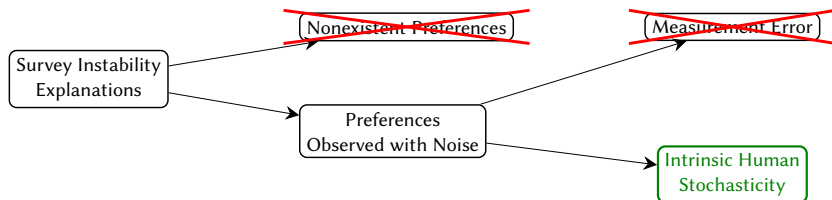
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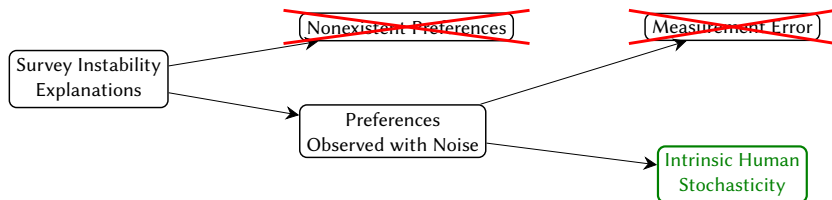
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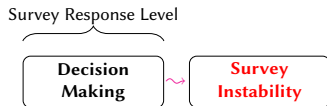


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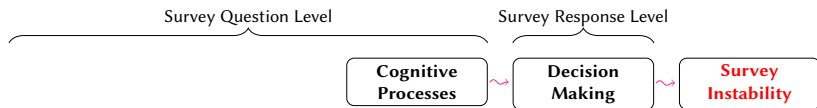
Sources of Stochasticity Leading to Survey Instability

**Survey
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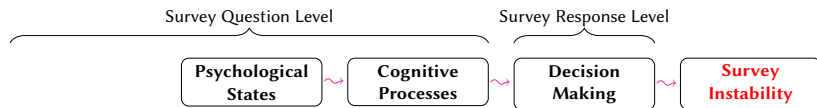
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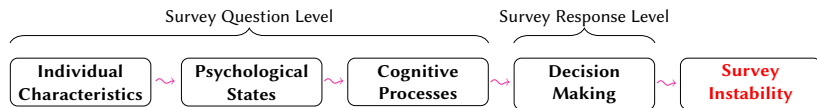
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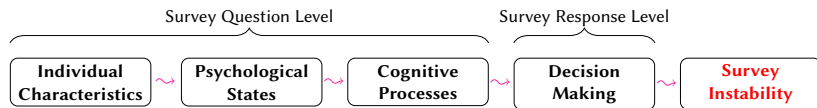
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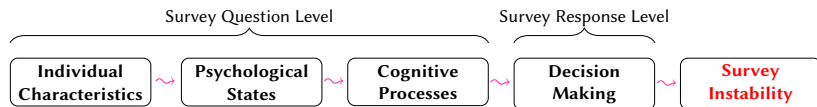


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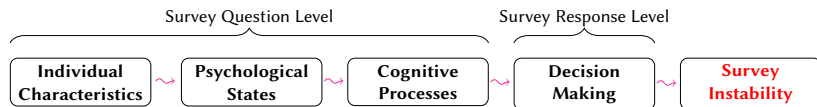
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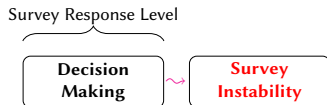
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C_t

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 - **Result:** **still** 40-60% stochastic! 🙅

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)

- Beliefs: $\theta \in \mathcal{C}(\mathcal{X} \times \mathcal{Y})$ (unknown)

- Cost

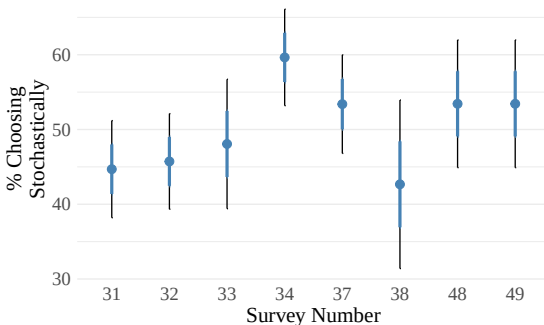
- History

C_t {

- Learning

- Exploration

Decision Making: Highly Stochastic!



Stochasticity Sources: Decision Making

- **Preference:** $\rho \in \{0, 1\}$ (unknown)
- **Belief:** $\pi = \Pr(\rho = 1)$ (known)
- **Choices:** C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- **How Beliefs become Choices**

$$C_t \begin{cases} = \mathbf{1}(\pi > 0.5) & \text{Expected Utility (deterministic)} & E(D) = 0 \\ = \mathbf{1}(\pi + \eta_t > 0.5) & \text{Random Utility (stochastic, } \eta_t \sim F) & E(D) > 0 \\ \sim \text{Bernoulli}(\pi) & \text{Probability Matching (stochastic)} & E(D) > 0 \end{cases}$$

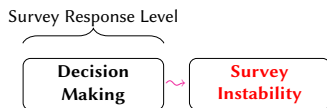
- **Literature:** Probability matching! (humans, bees, fish, pigeons...)
- **Experiments:**
 - **Tell (teach) respondents their beliefs (π)**
 - **Minimize:** η , uncertainty, indifference
 - **Result:** **still** 40-60% stochastic! 🙅

Stochasticity Sources: Cognitive Processing

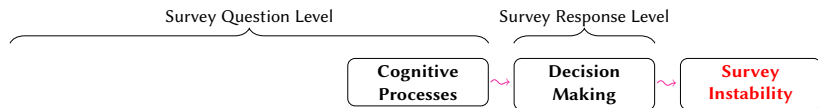
Stochasticity Sources: Cognitive Processing

**Survey
Instability**

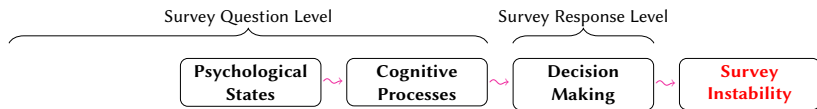
Stochasticity Sources: Cognitive Processing



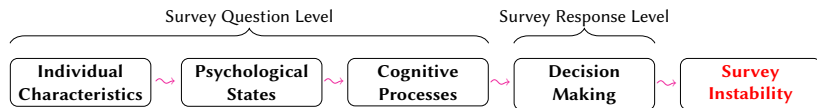
Stochasticity Sources: Cognitive Processing



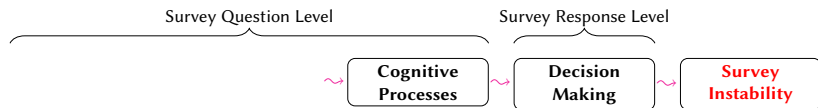
Stochasticity Sources: Cognitive Processing



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Stochasticity Sources: Cognitive Processing

What increases stochasticity (and thus instability)?

Stochasticity Sources: Cognitive Processing

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- High cognitive complexity: hard Qs; many As

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- Low time-on-task: for any given question

Stochasticity Sources: Cognitive Processing

What increases stochasticity (and thus instability)?

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- Differential cognitive processing: Q_1 vs Q_2

Stochasticity Sources: Cognitive Processing

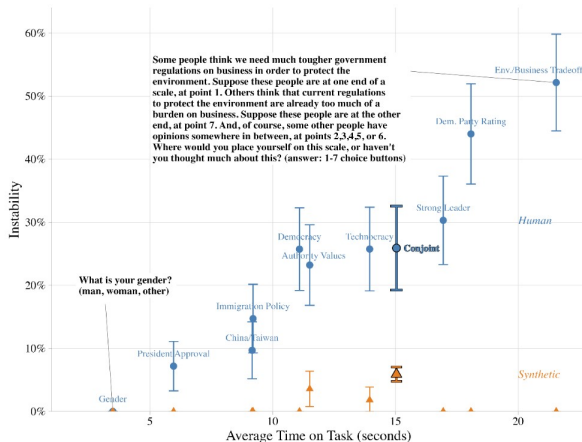
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Stochasticity Sources: Cognitive Processing

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- H
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Stochasticity Sources: Cognitive Processing

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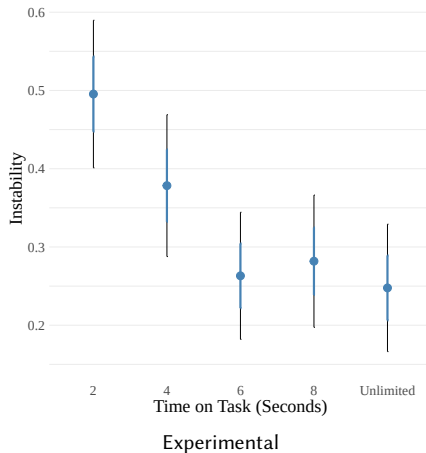
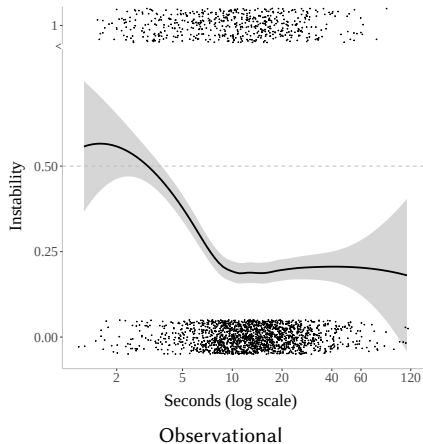
Stochasticity Sources: Cognitive Processing

What increases stochasticity (and thus instability)?

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Stochasticity Sources: Cognitive Processing

Limited time-on-task causes survey instability



Stochasticity Sources: Cognitive Processing

What increases stochasticity (and thus instability)?

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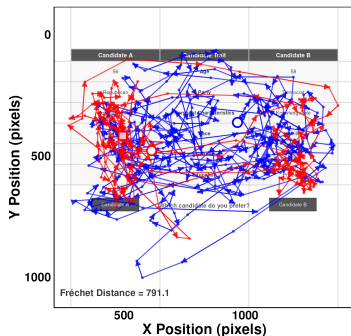
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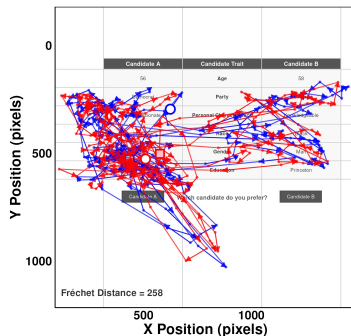
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Stochasticity Sources: Cognitive Processing

Stochasticity in Gaze Behavior

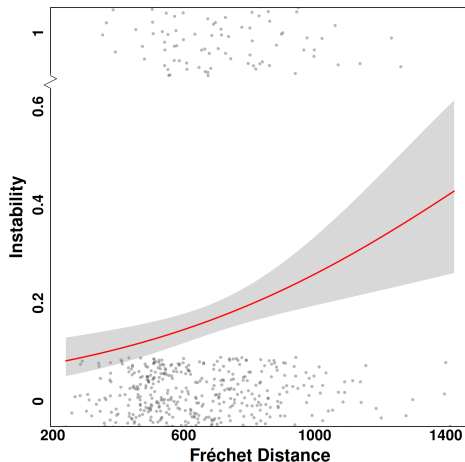


Large difference



Small difference

Stochastic Distance $\leadsto$ Survey Instability



Stochasticity Sources: Cognitive Processing

What increases stochasticity (and thus instability)?

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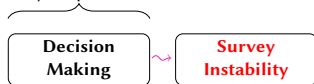
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Stochasticity Sources: Psychological States

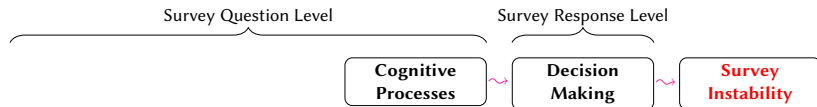
**Survey
Instability**

Stochasticity Sources: Psychological States

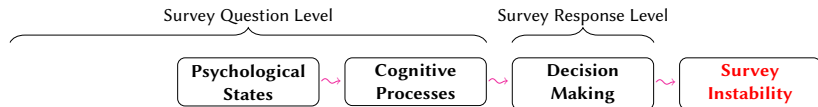
Survey Response Level



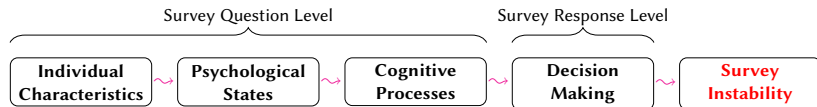
Stochasticity Sources: Psychological States



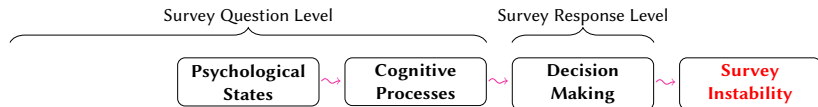
Stochasticity Sources: Psychological States



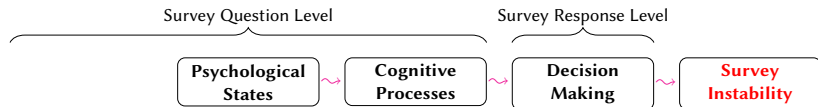
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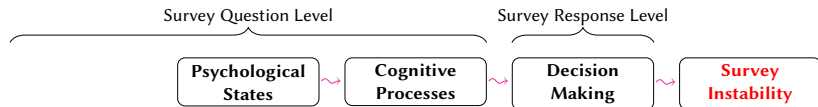


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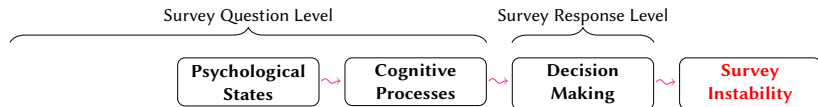
- Psychological States Summary

Stochasticity Sources: Psychological States



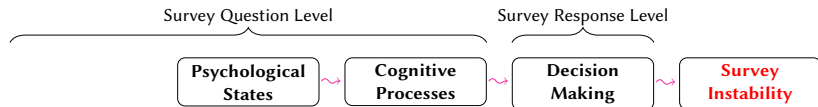
- **Psychological States Summary**
 - **Attention Checks** ruin surveys

Stochasticity Sources: Psychological States



- **Psychological States Summary**
 - **Attention Checks** ruin surveys
 - Use **Preoccupation** instead

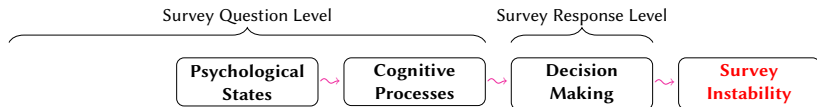
Stochasticity Sources: Psychological States



- **Psychological States Summary**
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Everyone comes to a new task (like this survey) with things on their minds. How preoccupied were you, just before you started this survey, with thoughts or worries about your partner, job, health, children, friends, money, or other concerns?

Stochasticity Sources: Psychological States



- **Psychological States Summary**
 - **Attention Checks** ruin surveys
 - Use **Preoccupation** instead

Everyone comes to a new task (like this survey) with things on their minds. How preoccupied were you, just before you started this survey, with thoughts or worries about your partner, job, health, children, friends, money, or other concerns?

{Very, Fairly, Slightly, Not at all} Preoccupied

So Who's to Blame for Survey Instability?

So Who's to Blame for Survey Instability?

Explanations:

So Who's to Blame for Survey Instability?

Explanations:

Survey Instability

~25%

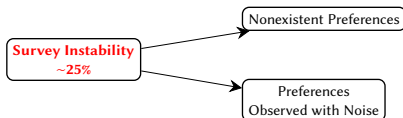
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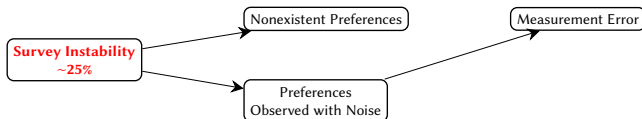
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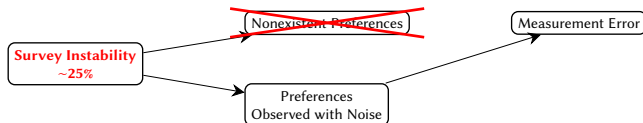
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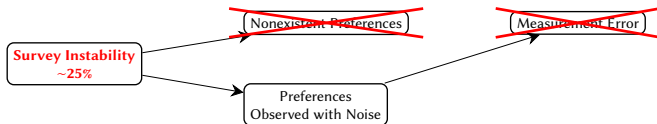
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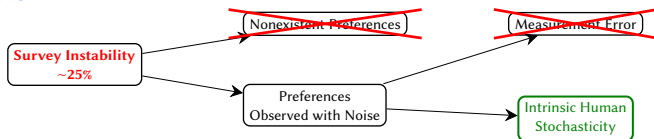
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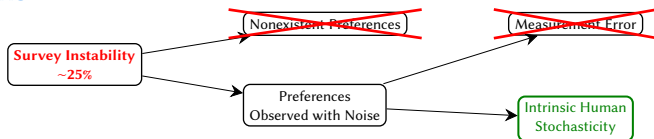
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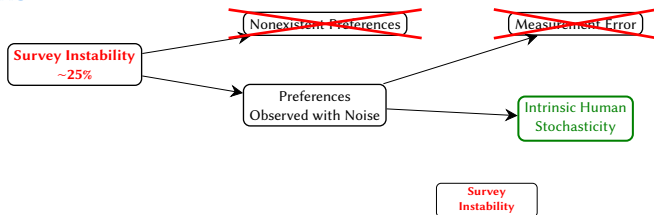
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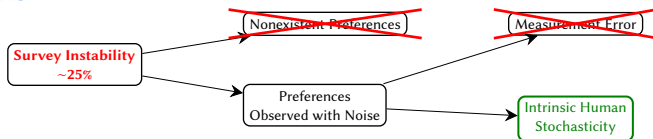
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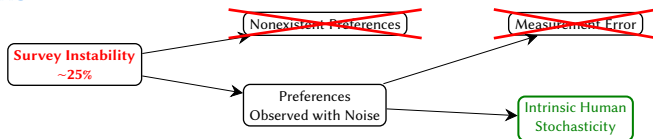


Sources:

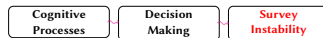


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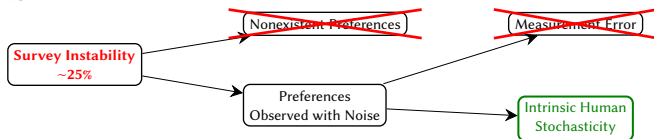


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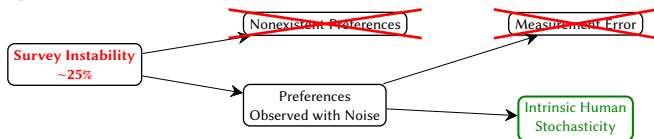


Sources:

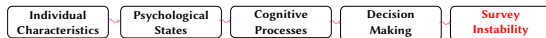


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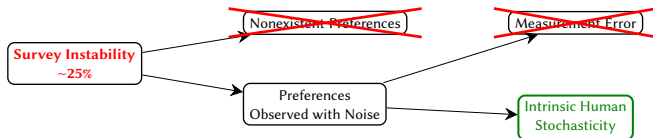


Sources:

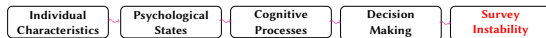


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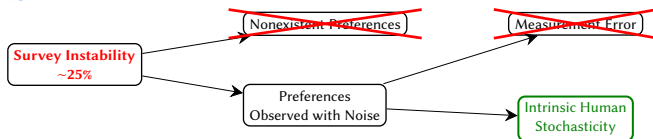
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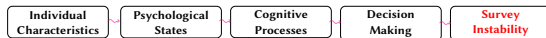
Implications:

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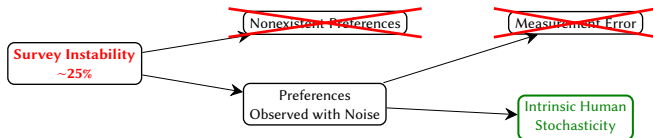


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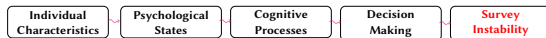
- Improve instruments:

So Who's to Blame for Survey Instability?

Explanations:



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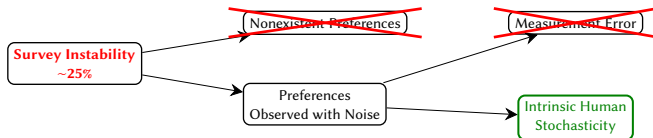


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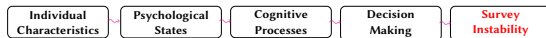
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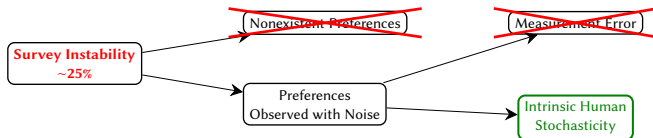


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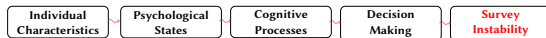
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So Who's to Blame for Survey Instability?

Explanations:



Sources:

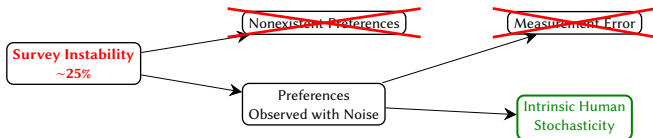


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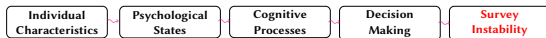
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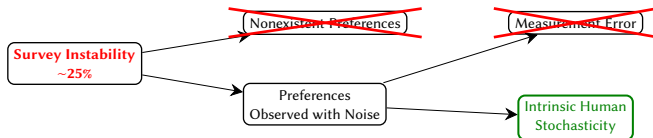


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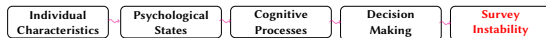
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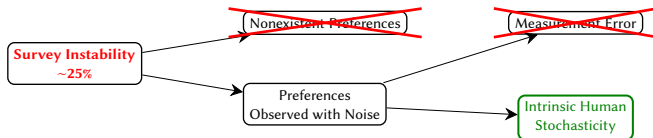


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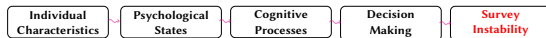
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Explanations:



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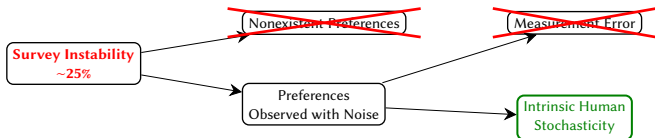


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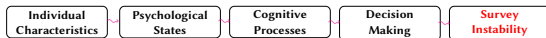
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So Who's to Blame for Survey Instability?

Explanations:



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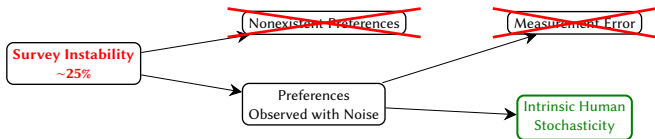


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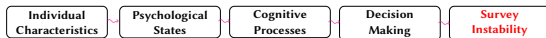
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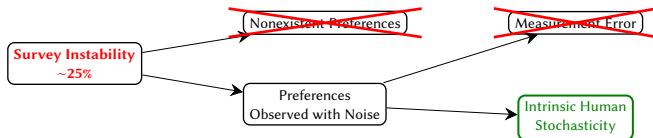


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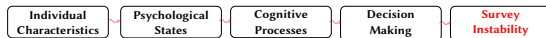
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So Who's to Blame for Survey Instability?

Explanations:



Sources:



Implications:

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